LAB MANUAL

Data Science

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Analyzing Monthly Sales Data for Central Tendencies

Objective:

- Compute the correlation and covariance between different manufacturing metrics.
- Interpret these measures to understand dependencies and trends.

Prerequisites

- Basic knowledge of statistics (mean, median, mode).

Requirements:

- Python installed (version 3.x recommended).
- NumPy and SciPy libraries installed (pip install numpy scipy).
- Jupyter Notebook (optional but recommended for running the lab).

Problem Statement:

Understanding the relationship between different variables in the manufacturing industry is crucial for optimizing processes and improving efficiency. Correlation and covariance help in identifying dependencies between key factors such as productivity, idle time, over time, and incentives.

Code:

```
# Import necessary libraries
import pandas as pd
import numpy as np
```

```
# Load dataset
df = pd.read_csv("garments_worker_productivity.csv")
```

```
# Select relevant numerical columns for analysis
selected_columns = ['targeted_productivity', 'smv', 'wip', 'over_time', 'incentive',
                    'idle_time', 'idle_men', 'no_of_style_change', 'no_of_workers', 'actual_productivity']
data = df[selected_columns]
```

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```python
# Compute covariance
cov_matrix = data.cov()

# Compute correlation
corr_matrix = data.corr()
```

```python
# Print results
print("Covariance Matrix:\n", cov_matrix)
print("Correlation Matrix:\n", corr_matrix)
```

Output:

```
Covariance Matrix:
                        targeted_productivity           smv           wip  \
targeted_productivity               0.009583     -0.074439  1.163078e+01
smv                                -0.074439    119.754046 -4.850380e+02
wip                                11.630780   -485.038033  3.376241e+06
over_time                         -29.030617  24732.539468  1.173852e+05
incentive                           0.513815     57.195571  8.478812e+03
idle_time                          -0.069899      7.908797 -8.076323e+02
idle_men                           -0.017222      3.788411 -3.834594e+02
no_of_style_change                 -0.008766      1.476655 -7.142011e+01
no_of_workers                      -0.183154    221.580535  5.257543e+02
actual_productivity                 0.007201     -0.233124  3.729937e+01

                          over_time     incentive    idle_time    idle_men  \
targeted_productivity -2.903062e+01      0.513815    -0.069899   -0.017222
smv                    2.473254e+04     57.195571     7.908797    3.788411
wip                    1.173852e+05   8478.812114  -807.632285 -383.459374
over_time              1.121462e+07  -2571.212375  1321.044885 -196.101555
```